Adiabatic Energy-Recycling Charge-Redistribution Array for Ultra-Low Power Massively Parallel Cross-Correlation

Shashank Bansal*, Pål Gunnar Hogganvik‡, Soumil Jain*, Gopabandu Hota§
Bouchaib Cherif†, Johannes Leugering¶ & Gert Cauwenberghs*

* *Dept. of Bioengineering, University of Califorina, San Diego, La Jolla, CA, US*
‡*Dept. of Informatics*, University of Oslo, Oslo, Norway
§Qernel Inc., Palo Alto, CA, US
†Northrop Grumman, Falls Church, VA, US
¶FZ Jülich, Jülich, Germany
*{shbansal, s3jain, gert}@ucsd.edu
‡paalgh@ifi.uio.no, §gopa@qernel.ai, †bouchaib.cherif@ngc.com, ¶j.leugering@fz-juelich.de

***Abstract*—We present a mixed-signal Compute-in-Memory (CIM) circuit for XNOR-and-Accumulate (XAC) that uses adiabatic charge redistribution to reduce switching loss and noise while recycling energy during computation. For bipolar 256-dimensional vectors in 22 nm fully depleted silicon-on-insulator (FD-SOI) simulations, the array core achieves an energy efficiency exceeding 1500 TOPS/W at 29.1 aJ per XAC and supports 8-bit quantization. The cell comprises 14 minimum-size transistors and two small unit capacitor, yielding a compact, energy-efficient, and high-throughput solution suitable for AI and signal-processing tasks in power- and thermal-constrained settings.**

***Index Terms*—adiabatic, compute-in-memory, Static random-access memory (SRAM), capacitive coupling**

I. Introduction

The rapid growth of artificial intelligence (AI) applications has intensified the demand for energy-efficient accelerators capable of large-scale computation [1], [2]. Modern neural models process and store vast datasets for inference, exacerbating the *data-movement problem*—where the time and energy of moving data dominate the overall system cost.

Compute-in-Memory (CIM) architectures have emerged as a promising solution by integrating computation directly within memory arrays [3]. By minimizing data transfer between memory and compute units, CIM significantly improves both energy efficiency and throughput. Early analog CIM implementations, such as *Kerneltron* [4]–[6], pioneered charge-domain correlation using charge injection onto a summing capacitor. However, these designs relied on nonlinear capacitive coupling and volatile charge storage, leading to signal decay and reduced computational fidelity over time. Their intrinsic nonlinearity and the charge-leakage behavior of dynamic storage elements limited precision and long-term stability.

In contrast, the mixed-signal CIM architecture presented in this work achieves high linearity and stability through fully differential, *capacitive coupling* controlled by static SRAM storage nodes. Binary weight values are stored in the 6T SRAM core through gate-charge redistribution paths, achieving near-zero static loss and eliminating the volatility typical of charge-injection memories. This approach enables highly repeatable computation with much greater signal fidelity than the Kerneltron-style devices, while retaining the intrinsic parallelism of charge-domain processing.

The proposed design employs a 14T–2C SRAM-based, capacitively coupled cell that performs XNOR-and-Accumulate (XAC) operations using *passive charge redistribution*—executing computation through charge sharing without active power consumption during the redistribution phase. An adiabatic "hot clock" (HC) generated by Energy Recovery Logic (ERL) [7], [8] gradually charges and discharges capacitive nodes, recycling a large fraction of switching energy and further minimizing dynamic loss. Finally, by removing the switched interface between the CIM array and the downstream ADC and instead directly sharing charge into the ADC capacitor network, the design suppresses kT/C-limited sampling noise and allows aggressive scaling of unit capacitors without compromising signal integrity. These innovations result in a compact, energy-efficient architecture capable of high-throughput matrix–vector multiplication (MVM) in power- and thermal-constrained AI systems, providing both the linearity and energy benefits that the earlier Kerneltron architectures could not achieve.

II. Architecture

Our CIM architecture employs a standard 6-transistor (6T) SRAM cell for local storage (Fig. 1). During the **PROGRAM** phase, complementary digital binary weights, denoted as $\texttt{WEIGHT}_+$ and $\texttt{WEIGHT}_-$, are programmed into the SRAM cell using bit-line drivers. During the **COMPUTE** phase, the circuit performs a signed XAC operation between the stored weights and the column-parallel, complementary binary-encoded analog inputs ($\texttt{IN}_+$ and $\texttt{IN}_-$). These analog inputs are differentially capacitively coupled onto two row-

(c) 2026 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works. Accepted manuscript of: S. Bansal et al., IEEE ISCAS 2026. DOI: 10.1109/ISCAS66217.2026.11562109

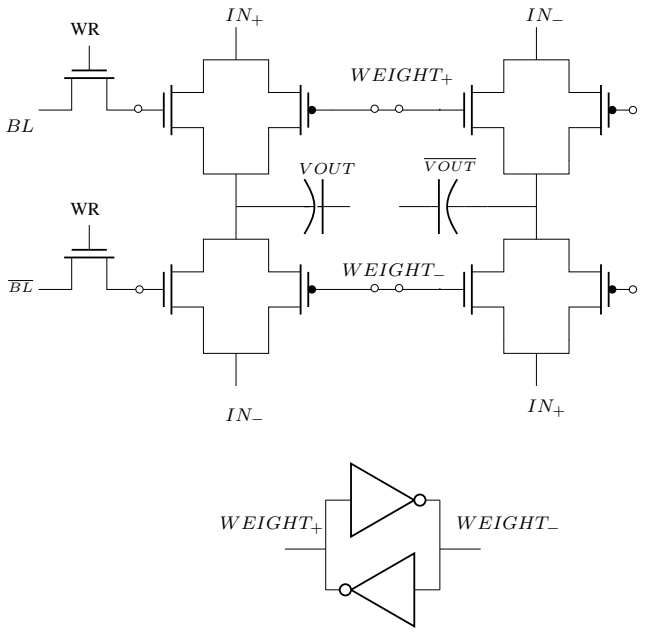


Fig. 1: 14T–2C SRAM-C unit cell.

parallel summing output nodes, V_{OUT} and $\overline{V}_{\text{OUT}}$ (Fig. 1). This configuration enables each row of cells to execute a fully parallel inner product between a stored template vector and a presented input vector, yielding an analog output linearly proportional to the correlation between the two vectors.

Each cell contributes charge onto the shared output lines through its coupling capacitor. The resulting output voltage after charge sharing can be expressed as

$$V_{\text{OUT}} = \frac{\sum_{k=1}^{N} C_{\text{c},k}\, V_{\text{IN},k}}{\sum_{k=1}^{N} C_{\text{c},k} + C_{\text{par}}}, \tag{1}$$

where $C_{\text{c},k}$ is the coupling capacitance of the k-th cell, $V_{\text{IN},k}$ is its applied analog input, N is the number of cells in the row, and C_{par} represents the total parasitic capacitance on the row line (including wiring and ADC input). Equation (1) shows that the output voltage naturally forms a weighted sum of the coupled input voltages, scaled by the capacitive division between the active coupling capacitors and the total load on the line. Thus, the "accumulate" operation arises purely from charge redistribution among capacitors, without any direct current summation.

To generate the analog input signals $\texttt{IN}_+$ and $\texttt{IN}_-$, we utilize the circuit shown in Fig. 2(a). It employs three switches to multiplex between three voltage levels: VDD (supply voltage), GND (ground), or HC (Hot Clock). Initially, $\texttt{IN}_+$ and $\texttt{IN}_-$ are held at either VDD or GND, ensuring a consistent starting point before computation begins.

When the COMPUTE signal transitions high, the circuit activates and generates the analog inputs based on the digital input data. Depending on the value of each digital input bit, the circuit dynamically routes one of the analog input signals to follow the HC signal for one clock cycle, while the complementary signal remains connected to GND, as shown in Fig. 2(b). Specifically:

- If the digital input bit is **'1'**: $\texttt{IN}_+$ is connected to HC, causing it to follow the adiabatic sinusoidal waveform of HC, while $\texttt{IN}_-$ remains connected to GND.
- If the digital input bit is **'0'**: $\texttt{IN}_-$ is connected to HC, and $\texttt{IN}_+$ remains connected to GND.

This operation results in differential analog input signals where one input is driven by the hot clock and the other remains at ground potential. The adiabatic modulation achieved through the HC signal minimizes energy dissipation by recycling charge, thereby contributing to the circuit's overall energy efficiency.

After coupling through the CIM array, the voltage difference on the output nodes represents the accumulated correlation between the input and stored weight vectors. This differential voltage can be expressed as

$$\Delta V_{\text{OUT}} \propto \frac{C_{\text{c}}}{C_{\text{row}}} \sum_{k=1}^{N} \alpha_k\, \Delta V_{\text{IN},k}, \tag{2}$$

where α_k denotes the effective sign (or phase) determined by the stored weight polarity, C_{row} is the total capacitance seen by the output node, and $\Delta V_{\text{IN},k}$ is the differential input voltage driven by the hot clock. Equation (2) shows that the resulting output voltage scales linearly with the total capacitive coupling, providing a direct analog representation of the accumulated result.

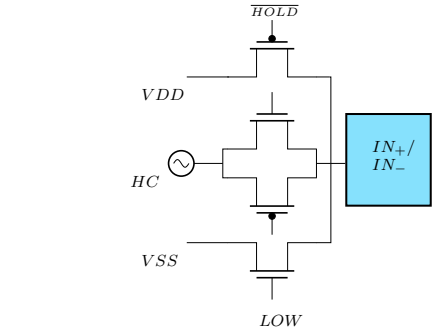
Finally, the output of each row connects directly to the input of a capacitive DAC within the successive-approximation ADC. The ADC senses this voltage through a passive capacitive divider ensuring continuous charge-domain transfer between the CIM array and ADC input without any sampling switches or kT/C noise penalties.

III. Results

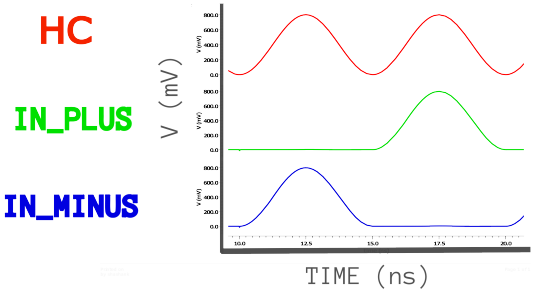
We simulated the proposed circuit in a 22 nm FDSOI process under typical conditions. The simulations show that each XAC operation consumes less than 30 aJ of energy—more than $30\times$ lower than any other reported XAC circuit, including those implemented in more advanced process nodes (see Table II). This substantial reduction highlights the effectiveness of our design in achieving ultra-low-power computation.

The higher power consumption observed during programming operations comes from the WR line and driving the differential bit lines to supply the weights. However, these write operations occur infrequently compared to compute operations. In matrix-vector multiplication (MVM), the same weight matrix (e.g., the DNN coefficients) is multiplied with multiple input vectors (e.g., activations or sensor data). As a result, the energy required for programming the weights is amortized over many compute cycles and thus has a negligible impact on overall energy efficiency.

A key innovation of our design is the use of passive charge redistribution between the capacitively coupled CIM array and the Analog-to-Digital Converter (ADC). By exploiting direct



(a)



(b)

Fig. 2: (a) Energy Recovery Logic (ERL) block diagram. (b) Hot clock and the corresponding generated differential analog input signals. In traditional CMOS, energy is dissipated during the charging and discharging of capacitive loads at each output node, resulting in CV^2 losses. Adiabatic circuits use gradual charging and discharging, minimizing resistive losses and enabling partial energy recovery back into the power supply.

charge sharing, we eliminate the need for any active switching components between these two stages. This simplification removes sampling noise—a fundamental limitation in conventional switched-capacitor circuits caused by kT/C thermal noise during capacitor charging and discharging. The absence of sampling noise also allows a significant reduction in the size of the unit capacitors without degrading signal integrity or the accuracy of the XAC operation. Smaller capacitors not only occupy less silicon area but also require less charge to operate, resulting in further power savings.

The shared-capacitance interface between the CIM and ADC operates through a passive divider, enabling seamless voltage transfer to the SAR input while avoiding any sampling switches or additional conversion energy. This direct coupling ensures that the ADC power overhead remains minimal and the overall system retains its adiabatic energy-recycling efficiency.

Moreover, because all charge redistribution occurs through well-matched capacitive ratios, the accumulated output voltage scales linearly with the number of active cells, and parasitic effects manifest only as a small, predictable attenuation factor that can be pre-calibrated or absorbed into the ADC gain.

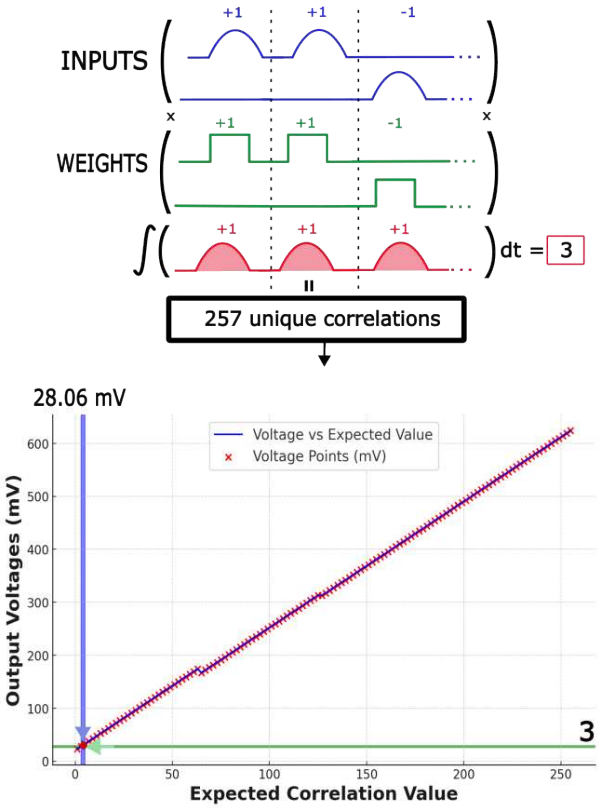


Fig. 3: Output voltage versus expected correlation following an XAC operation.

IV. Performance Metrics

A. XNOR-and-Accumulate (XAC)

Fig. 3 illustrates the XAC process performed by the proposed CIM array using bipolar input and weight vectors. Each 256-dimensional computation yields one of 257 possible correlation values, producing an analog output voltage linearly proportional to the inner product of the two vectors. The output distribution thus spans discrete voltage levels corresponding to each possible correlation state.

As shown in Fig. 3, the measured output exhibits excellent linearity across the full range, ensuring that the analog voltages can be accurately quantized by the ADC with minimal distortion or code mapping error.

B. Energy Efficiency

Energy efficiency was evaluated through post-simulation analysis at a 200 MHz hot-clock frequency. The total integrated power for a single 1×256 row was $E_{\text{row,total}} = 60.29\,\text{fJ} - 0.563\,\text{fJ} = 59.73\,\text{fJ}$. Dividing across eight compute cycles gives $E_{\text{row,cycle}} = 59.73/8 = 7.46\,\text{fJ}$ per cycle. Since each row contains 256 cells performing one XAC per cycle, the per-cell energy is

$$E_{\text{cell,XAC}} = \frac{7.46\,\text{fJ}}{256} = \mathbf{29.1\,aJ/XAC.}$$

These results, summarized in Table I, represent only the array core energy. Peripheral components such as ADCs and

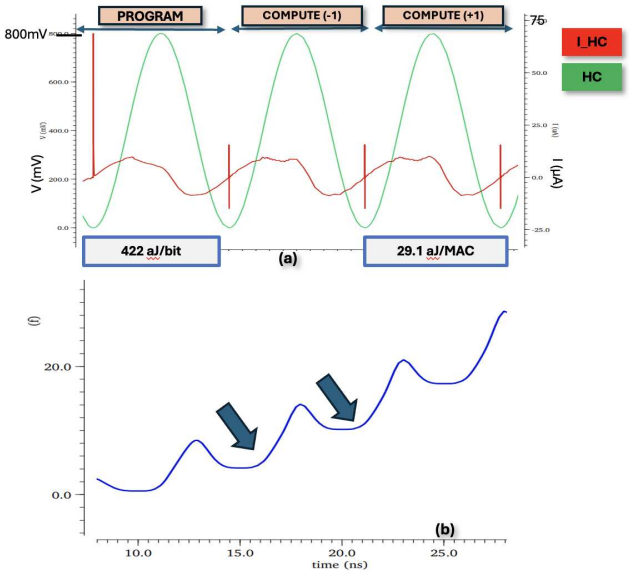


Fig. 4: (a) Simulated energy consumption of a 1×256 SRAM-CIM array in 22 nm FDSOI. (b) Integrated instantaneous power—blue arrows indicate adiabatic energy recovery across compute cycles.

line drivers are excluded here. In practical systems, low-power or adiabatic peripheral circuits can further reduce total energy by recycling capacitive drive energy and offsetting I/O overheads.

TABLE I: Energy calculation summary per compute cycle.

Metric	Energy (fJ)
Per row (8 cycles)	59.73
Per row per cycle	7.46
Per CIM cell (per XAC)	**0.029**

Table II compares this work against recent SRAM-based CIM architectures and the Kerneltron for historical context. The proposed 22 nm adiabatic SRAM-CIM achieves over **1500 TOPS/W** efficiency and 29 aJ per XAC—surpassing both earlier charge-injection CIMs (e.g., Kerneltron) and recent advanced-node designs.

TABLE II: Comparison with recent SRAM-CIM architectures.

	JSSC'21 [9]	ISSCC'2 [10]	JSSC'22 [11]	ISCAS'2 [12]	ISSCC'0 [8]	This work
Technology	7 nm	28 nm	16 nm	40 nm	350 nm	**22 nm**
Supply (V)	0.8	0.8	0.8	0.7–0.9	1.65	**0.7–0.9**
Op. Mode	AIM	AIM	AIM	AIM	AIM	**AIM**
Energy Eff. [TOPS/W]	351	22.75	30	282–607	960	**>1500**

AIM → Analog-In-Memory; Values as reported in respective papers. The Kerneltron (350 nm) included for historical comparison.

V. Discussion

In this paper, we introduced a fast and efficient mixed-signal Compute-In-Memory (CIM) architecture optimized for XNOR-and-Accumulate (XAC) operations. Our design addresses the energy and performance bottlenecks in modern AI workloads caused by the *data-movement problem* inherent to von Neumann architectures. By embedding computation within the memory array, the proposed CIM framework reduces data transfer, thereby enhancing both energy efficiency and computational throughput.

The presented architecture also represents a modern evolution of earlier charge-domain CIM designs such as the *Kerneltron* family [4]–[6], [8]. While the Kerneltron pioneered charge-injection correlation, its nonlinear capacitive coupling and volatile charge storage limited precision and signal stability. In contrast, the proposed 14T–2C SRAM-based CIM uses linear capacitive coupling controlled by static SRAM storage nodes, achieving high fidelity, repeatability, and near-zero static loss. This linearity eliminates the charge-decay and signal-dependent nonlinearity of injection-based designs. Combined with adiabatic energy recovery, the circuit performs low-loss charge redistribution and substantially extends energy efficiency beyond prior dynamic CIM approaches.

Despite these advantages, several challenges remain for practical large-scale deployment: (a) the *complexity of adiabatic circuit design* at high frequency, requiring optimization for standard CMOS; (b) efficient *integration with digital systems*, which depends on precise, low-power ADCs for seamless charge-domain conversion; (c) maintaining performance during *scaling to larger arrays or higher precision*, which may demand hierarchical or modular architectures to manage parasitic capacitance; and (d) mitigating *analog noise and mismatch effects*, especially at the ADC interface, through calibration and adaptive compensation.

Managing thermal effects in dense arrays also remains important for long-term reliability. Future work will include implementing high-efficiency ADCs, validating fabricated prototypes, and exploring multi-bit operation. Appropriate selection of ADC resolution is critical for optimizing the precision-energy tradeoff in analog in-memory computing, as explored in [13]. The architecture's energy efficiency and compact design also make it suitable for neuromorphic and event-driven computing applications [14]. Overall, by combining linear capacitive coupling, static SRAM control, and adiabatic energy recovery, this work extends the Kerneltron concept into a stable, high-fidelity, and energy-recovering CIM platform for next-generation AI and signal-processing systems.

References

[1] N. R. Shanbhag and S. K. Roy, "Benchmarking In-Memory Computing Architectures," *IEEE Open Journal of the Solid-State Circuits Society*, vol. 2, pp. 288–300, 2022. Conference Name: IEEE Open Journal of the Solid-State Circuits Society.

[2] S. Bavikadi, P. R. Sutradhar, K. N. Khasawneh, A. Ganguly, and S. M. Pudukotai Dinakarrao, "A Review of In-Memory Computing Architectures for Machine Learning Applications," in *Proceedings of the 2020 on Great Lakes Symposium on VLSI*, GLSVLSI '20, (New

York, NY, USA), pp. 89–94, Association for Computing Machinery, Sept. 2020.

[3] R. Ronen, A. Eliahu, O. Leitersdorf, N. Peled, K. Korgaonkar, A. Chattopadhyay, B. Perach, and S. Kvatinsky, "The Bitlet Model: A Parameterized Analytical Model to Compare PIM and CPU Systems," *J. Emerg. Technol. Comput. Syst.*, vol. 18, pp. 43:1–43:29, Mar. 2022.

[4] R. Genov and G. Cauwenberghs, "Kerneltron: support vector "machine" in silicon," *IEEE Transactions on Neural Networks*, vol. 14, pp. 1426–1434, Sept. 2003.

[5] R. Karakiewicz, R. Genov, and G. Cauwenberghs, "1.1 TMACS/mW Load-Balanced Resonant Charge-Recycling Array Processor," in *2007 IEEE Custom Integrated Circuits Conference*, pp. 603–606, Sept. 2007. ISSN: 2152-3630.

[6] R. Karakiewicz, R. Genov, A. Abbas, and G. Cauwenberghs, "175 GMACS/mW Charge-Mode Adiabatic Mixed-Signal Array Processor," in *2006 Symposium on VLSI Circuits, 2006. Digest of Technical Papers.*, (Honolulu, HI, USA), pp. 100–101, IEEE, 2006.

[7] Y. Moon and D.-K. Jeong, "An efficient charge recovery logic circuit," *IEEE Journal of Solid-State Circuits*, vol. 31, pp. 514–522, Apr. 1996. Conference Name: IEEE Journal of Solid-State Circuits.

[8] R. Karakiewicz, R. Genov, and G. Cauwenberghs, "480-GMACS/mW Resonant Adiabatic Mixed-Signal Processor Array for Charge-Based Pattern Recognition," *IEEE Journal of Solid-State Circuits*, vol. 42, pp. 2573–2584, Nov. 2007.

[9] M. E. Sinangil, B. Erbagci, R. Naous, K. Akarvardar, D. Sun, W.-S. Khwa, H.-J. Liao, Y. Wang, and J. Chang, "A 7-nm Compute-in-Memory SRAM Macro Supporting Multi-Bit Input, Weight and Output and Achieving 351 TOPS/W and 372.4 GOPS," *IEEE Journal of Solid-State Circuits*, vol. 56, pp. 188–198, Jan. 2021.

[10] J.-W. Su, Y.-C. Chou, R. Liu, T.-W. Liu, P.-J. Lu, P.-C. Wu, Y.-L. Chung, L.-Y. Hung, J.-S. Ren, T. Pan, S.-H. Li, S.-C. Chang, S.-S. Sheu, W.-C. Lo, C.-I. Wu, X. Si, C.-C. Lo, R.-S. Liu, C.-C. Hsieh, K.-T. Tang, and M.-F. Chang, "16.3 A 28nm 384kb 6T-SRAM Computation-in-Memory Macro with 8b Precision for AI Edge Chips," in *2021 IEEE International Solid-State Circuits Conference (ISSCC)*, vol. 64, pp. 250–252, Feb. 2021. ISSN: 2376-8606.

[11] H. Jia, M. Ozatay, Y. Tang, H. Valavi, R. Pathak, J. Lee, and N. Verma, "Scalable and Programmable Neural Network Inference Accelerator Based on In-Memory Computing," *IEEE Journal of Solid-State Circuits*, vol. 57, pp. 198–211, Jan. 2022.

[12] Q. Zang, W. L. Goh, L. Lu, C. Yu, J. Mu, T. T.-H. Kim, B. Kim, D. Li, and A. T. Do, "282-to-607 TOPS/W, 7T-SRAM Based CiM with Reconfigurable Column SAR ADC for Neural Network Processing," in *2023 IEEE International Symposium on Circuits and Systems (ISCAS)*, pp. 1–5, May 2023. ISSN: 2158-1525.

[13] J. Leugering *et al.*, "Where to Cut: Efficient ADC Quantization for Analog in-Memory Computing with Discrete Values," in *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*, pp. 3188–3195, IEEE, 2025.

[14] G. Hota *et al.*, "Reconfigurable Event-Driven Spiking Neuromorphic Computing near High-Bandwidth Memory," in *2023 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, pp. 222–225, IEEE, 2023.